

SOFTWARE ENGINEERING
LAB 1
PROBLEM STATEMENT #59
COMMUNITY LOST AND FOUND MATCHING NETWORK

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Community Lost & Found Matching Network

Architecture Selection: Layered Architecture

Architectural Choice:

The system is divided into separate layers based on their responsibilities. The presentation layer handles user interaction, the business/application layer handles item management, matching, notifications and ownership verification, and the data layer manages the Lost & Found database.

- Separation of Responsibilities:
Layered architecture separates different system functions into independent components. For example, item registration, item matching and ownership verification can be handled separately. This makes the system easier to understand, maintain and modify.
- Suitable for the System Requirements:
The Lost & Found system follows a clear flow from user input to processing and data storage. Users submit lost or found item details, the system validates and matches the reports, and the relevant information is stored in the database. Layered architecture fits this structured workflow.

Security Advantage:

Sensitive information such as finder contact details can be protected by keeping ownership verification in the business layer. Contact details are revealed only after successful verification, reducing unauthorized access to user information.

Performance Benefit:

The matching functionality can be optimized separately within the business layer. Since the Matching Engine handles category, color/tag and geographical matching, it can be improved without changing the user interface or database layer. This helps the system provide faster matching results.

COMMUNITY LOST AND FOUND

